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C-H Activation Hot Paper

Asymmetric Ruthenium-Catalyzed C–H Activation by a Versatile Chiral-Amide-Directing Strategy

*Wenkun Chen, Jijun Jiang, and Jun Wang**

Abstract: A versatile and readily available chiral amide directing group has been developed for the ruthenium-(II)-catalyzed asymmetric C–H activation. Asymmetric C–H activation of the related chiral benzamides with various olefins, aldehydes and propargylic alcohols has been accomplished with high stereoselectivities, affording a series of chiral products including 3,4-dihydroisocoumarins (up to 96 % ee), isocoumarins (up to 92 % ee), phthalides (up to 99 % ee), chiral bicyclo[2.2.1]heptanes (>20:1 dr), 4-alkylidene-3,4-dihydroisocoumarins (up to 97 % ee) and allenes (>20:1 dr). Importantly, our methodologies enabled concise syntheses of many biologically active compounds and natural products (e.g., Montroumarin, Cyclosporone E, Cyclosporone Q, Concentricolide, Chuangxinol, and Eleutherol).

Introduction

Transition-metal-catalyzed, directing-group assisted asymmetric C–H activation enables enantioselective constructions of diverse valuable chiral compounds from cheap fundamental chemical feedstocks.[1] It proceeds in a highly atom- and step-economic fashion, featuring a sequential process of cleavage of an inert C–H bond and then functionalization. It is well known that palladium, rhodium, and ruthenium are currently three most widely used transition metals to catalyze C–H activation. However, in terms of asymmetric C–H activation, although palladium-[1h,2] and rhodium-catalysis[1a,h,3] have achieved lots of progress, the ruthenium-catalysis[4] remains in its infancy.

Given the excellent catalytic performance of $[RuCl_2(p\text{-cymene})]_2$ in C–H activation,[5] which was first revealed by Oi and Inoue,[6] three strategies are reasonably deduced to make the corresponding reactions asymmetric.[4] The first one is the usage of a chiral directing group. In 2019, Cui[7a] and Wang[7b] independently reported an intramolecular hydroarylation of alkenes assisted by a chiral transient directing group. Very recently, Gandon and Sahoo reported an asymmetric two-fold C – H activation/annulation process assisted by a chiral sulfoximine directing group.[8] The second is the employment of a chiral carboxylic acid, which was elegantly demonstrated by Ackermann,[9] Shi,[10] Yoshino and Matsunaga[11] from 2021. The third is to replace p-cymene in $[RuCl_2(p\text{-cymene})]_2$ with a chiral benzene ligand, which was first revealed by Wang in 2022.[12] Besides, an asymmetric relay catalysis involving ruthenium-catalyzed C–H activation followed by a chiral amine-promoted asymmetric intramolecular Michael addition was also reported by Dethe.[13]

It is noteworthy that, despite its high potential for achieving ruthenium-catalyzed asymmetric C–H activation, the strategy of utilizing chiral directing group(DG*) remains largely overlooked by the community.[14] And the chiral directing group with wide applications is still missing. Herein, a broadly useful chiral amide directing group derived from commercially available and cheap (*R*)-α,α-diphenylprolinol is presented (Scheme 1). Assisted by it, asymmetric C–H activation of chiral benzamide with olefins, aldehydes, ketones, and propargylic alcohols have been accomplished with high diastereoselectivities, affording various chiral products with diverse chiral elements including 3,4-dihydroisocoumarins (center chirality, up to 96 % ee), 4-alkylidene-3,4-dihydroisocoumarins (center chirality, up to 97 % ee), isocoumarins (C–N axial chirality, up to 92 % ee), phthalides (center chirality, up to 99 % ee), chiral bicyclo[2.2.1]heptanes (center chirality, >20:1 dr) and allenes (axial chirality, >20:1 dr). It is worth noting that our methodologies can enable concise syntheses of several biologically active compounds and natural products (e.g., Montroumarin, Cyclosporone E, Cyclosporone Q, Concentricolide, Chuangxinol, and Eleutherol).

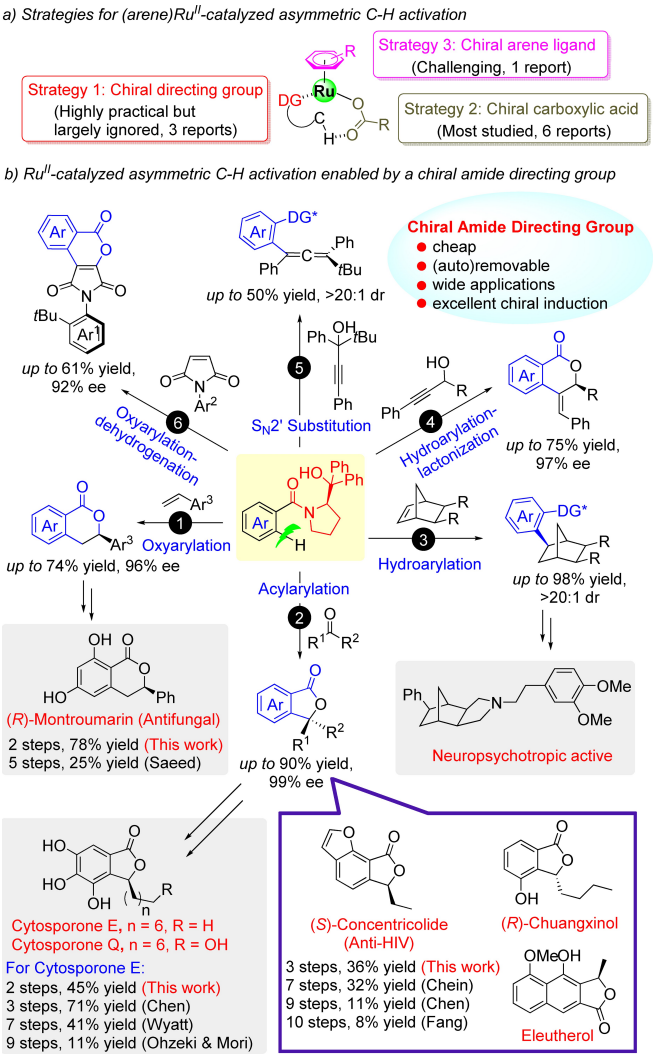
Results and Discussion

Reaction with Alkenes

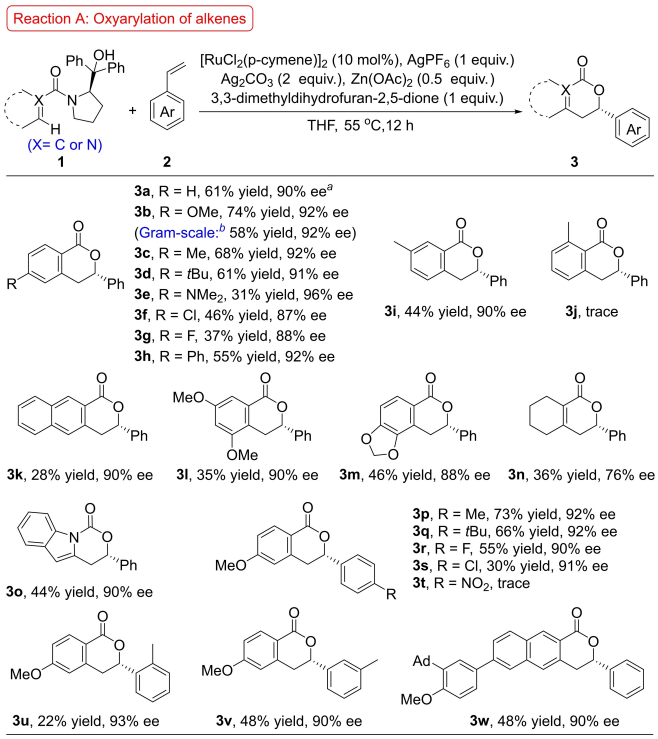
Chiral 3,4-dihydroisocoumarins widely exist in bioactive compounds and natural products.[15] Although several catalytic asymmetric syntheses of 3,4-dihydroisocoumarins have been reported,[16] the employed substrates or chiral catalysts are normally not easily available. In recent years, transition-metal catalyzed C–H activation turns out to be a straightforward strategy to synthesize various 3,4-dihydroisocoumarins by reaction of benzoic acids[17] or amides[18] with epoxides or alkenes, which are all facilely available. However, the related asymmetric catalysis is challenging and has not been achieved yet. Herein, we present the first example of

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Scheme 1. Development and synthetic applications of the ruthenium-(II)-catalyzed asymmetric C–H activation enabled by a robust chiral amide directing group.



Scheme 2. Reaction of chiral amides with alkenes. Reaction conditions: under N_2 atmosphere, amide **1** (0.2 mmol), olefin **2** (0.4 mmol), $[RuCl_2(p\text{-cymene})]_2$ (10 mol%), $AgPF_6$ (1 equiv.), Ag_2CO_3 (2 equiv.), $Zn(OAc)_2$ (0.5 equiv.), 3,3-dimethyldihydrofuran-2,5-dione (1 equiv.), THF (0.2 mL), 55 °C, and 12 h. [a] The absolute configuration was assigned to be *S* by comparison of its optical rotation with the reported value (see Supporting Information). [b] 3.0 mmol of **1b** was used.

asymmetric synthesis of 3,4-dihydroisocoumarins by a chiral amide-directed, ruthenium(II)-catalyzed asymmetric C–H activation.

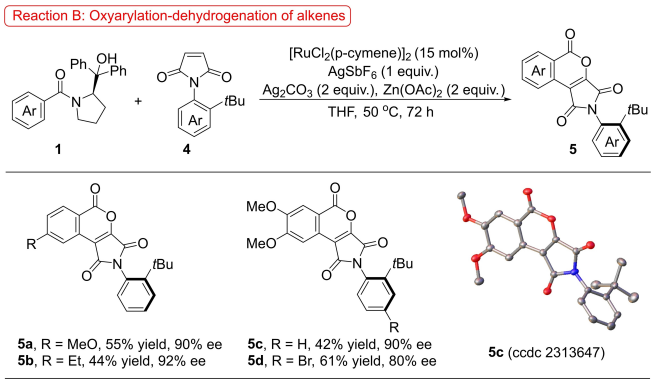
Under the optimized reaction conditions indicated in Scheme 2 (for reaction condition optimizations, see Supporting Information), various chiral amides **1** derived from (*R*)-α,α-diphenylprolinol were subjected to react with substituted styrenes, giving chiral 3-substituted 3,4-dihydroisocoumarins in moderate to good yields with high ee values. Benzamides bearing electron-withdrawing or electron-donating substituents were all tolerable (**3a–h**). The reaction of the chiral amide **1b** with styrene could be conducted on gram scale (3.0 mmol), affording **3b** in 58% yield with 92% ee. The product **3i** was obtained solely when 3-methyl substituted benzamide was tested. 2-Methyl substituted benzamide **1j** was unreactive. 2-Naphthoic acid derived and two multi-substituted benzamides were also suitable substrates for this transformation (**3k–m**). Cyclohexene-1-carboxamide afforded product in reasonable yield with 76% ee (**3n**). The indole derived product **3o** was also obtained in 44% yield with 90% ee. With 4-MeO substituted benzamide as model substrate, various substituted styrenes were then evaluated. All gave reasonable yields and high ee values (**3p–v**). Additionally, by this protocol, Adapalene, a retinoid drug for the treatment of acne, could be smoothly converted to the corresponding 3,4-dihydroisocoumarin **3w** in 48% yield with 90% ee. However, the reaction of **3a** with aliphatic alkenes such as 1-hexene was unsuccessful.

When N-aryl maleimides were tested, interestingly, isocoumarins **5** were generated instead of 3,4-dihydroisocoumarins, possibly by a sequential oxyarylation-dehydrogenation process (Scheme 3). And silver carbonate should play as the oxidant. To our delight, when N-2-^{t}Bu-phenyl maleimides were employed, isocoumarins **5a–d** with C–N axial chirality were successfully obtained in 42–61% yield with 80–92% ee under slightly modified reaction conditions.

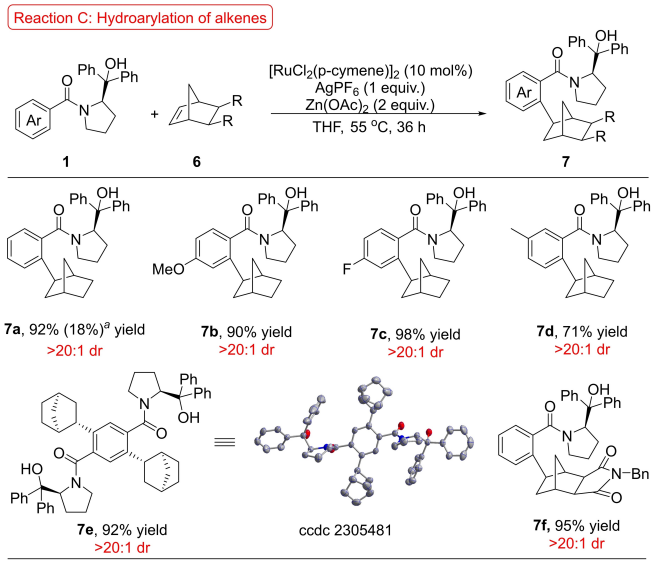
Furthermore, the chiral benzamide **1a** was also attempted to react with norbornene **6**.[19] Unexpectedly, the hydroarylation product **7a** was obtained (92% yield, >20:1 dr) instead of annulation products, which was likely due to the steric reasons (Scheme 4). Chiral amides with either electron-donating or electron-withdrawing substituents are good substrates (**7b–c**). High regioselectivity was observed for 3-methyl benzamide (**7d**). Dual C–H activation/alkylation was possible when the para-di-amide was

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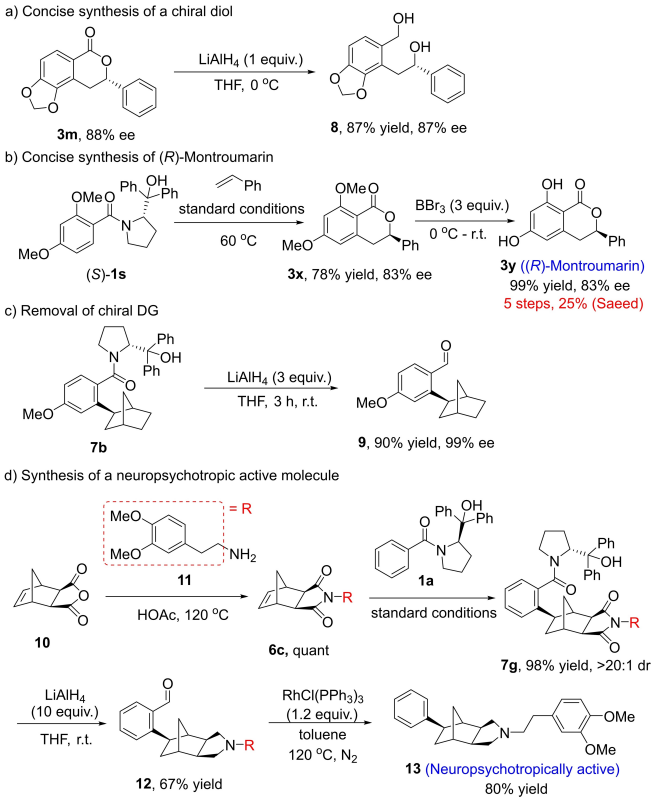
Scheme 3. Reaction of chiral amides with N-aryl maleimides. Reaction conditions: under N_2 atmosphere, amide **1** (0.2 mmol), maleimide **4** (0.1 mmol), $[RuCl_2(p\text{-cymene})]_2$ (15 mol%), $AgSbF_6$ (1 equiv.), Ag_2CO_3 (2 equiv.), $Zn(OAc)_2$ (2 equiv.), THF (0.1 mL), 50 °C and 72 h.



Scheme 4. Reaction of chiral amide with norbornene. Reaction conditions: under N_2 atmosphere, amide **1** (0.2 mmol), norbornene **6** (0.4 mmol), $[RuCl_2(p\text{-cymene})]_2$ (10 mol%), $AgPF_6$ (1 equiv.), $Zn(OAc)_2$ (2 equiv.), THF (0.2 mL), 55 °C, 36 h. [a] Under the reaction conditions of Reaction A.

applied, affording the corresponding product **7e** (92% yield, >20:1 dr). Its structure was unambiguously confirmed by single-crystal X-ray diffraction analysis. *N*-Benzylsuccinimide-fused norbornene was also reactive, leading to the product **7f** in 95% yield with >20:1 dr. However, when 7-azabenzonorbornadienes and 7-oxabenzonorbornadiene were employed, no desired products were observed.

As shown in Scheme 5, the utilities of this methodology was illustrated. First, the chiral diol **8** could be prepared in high yield without loss of enantiopurity by reduction of 3,4-dihydroisocoumarin **3m** with $LiAlH_4$ (Scheme 5a). Second, with our method as the key step, Montroumarin,[20] a natural product isolated from Montrouziera sphaeroidea, could be concisely synthesized via two steps. It includes the preparation of 3,4-dihydroisocoumarin **3x** from chiral amide (*S*)-**1s**



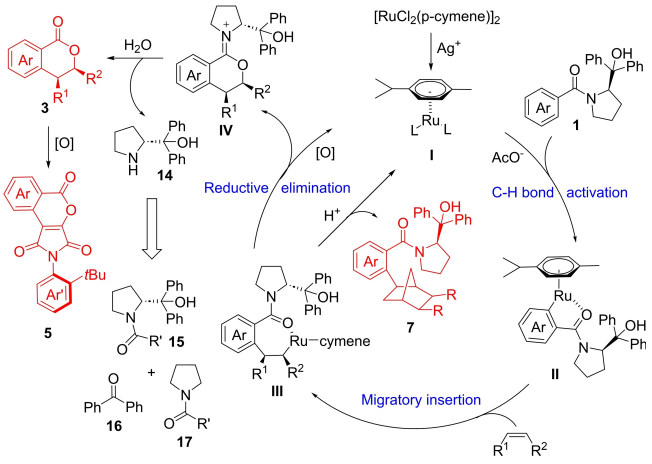
Scheme 5. Synthetic applications.

and styrene (78% yield, 83% ee), and removal of methyl with BBr_3 (99% yield, 83% ee) (Scheme 5b). In contrast, Saeed[21] took 5 steps to synthesize (*S*)-Montroumarin (25% yield) starting from 3,5-dimethoxyhomophthalic acid. Third, unexpectedly, the product **7b** could be converted to the chiral aryl aldehyde **9** by the treatment of $LiAlH_4$ (Scheme 5c). It was reasoned that the hydroxy of prolinol should coordinate to aluminum of the tetrahedral intermediate generated from the hydride addition to amide, which played the similar role as methoxy of Weinreb amide did. Upon workup, aldehyde was thus released. Remarkably, by taking advantage of this methodology, asymmetric synthesis of a neuropsychotropically active molecule **13**[22] was conveniently synthesized. As depicted in Scheme 5d, the synthesis began with the reaction of cis-5-norbornene-exo-2,3-dicarboxylic anhydride (**10**) with 3,4-di-*O*-methyldopamine (**11**) to give the *exo*-norbornene-2,3-dicarboximide **6c**. It was reacted with chiral amide **1a** to afford hydroarylation product **7g** in 98% yield with >20:1 dr. Reduction with excessive amount of $LiAlH_4$ led to the aldehyde **12** in 67% yield. Finally, removal of formyl by using Tsuji-Wilkinson decarbonylation method provided the targeted compound **13** in 80% yield.

A plausible reaction mechanism is proposed (Scheme 6). First, dechlorination of $[Ru(p\text{-cymene})Cl_2]_2$ by silver salt generates the catalytically active species **I**. Then the chiral amide **1** coordinates to ruthenium. With the assistance of acetate anion, the *ortho* C–H bond is cleaved through a concerted metalation-deprotonation (CMD) mechanism to

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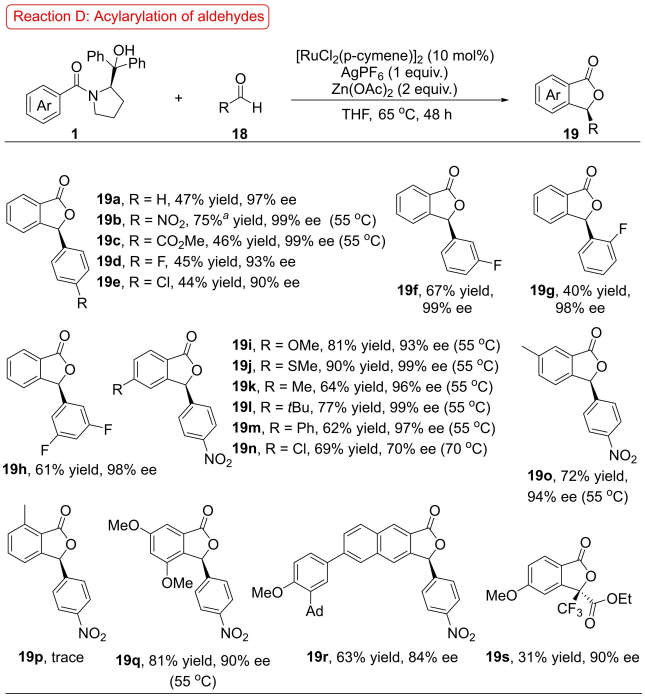
Scheme 6. Proposed reaction mechanisms for Reactions A–C.

give the ruthenacycle **II**. Coordination of olefins followed by migratory insertion gives the intermediate **III**. After reductive elimination and hydrolysis of the iminium intermediate **IV**, the chiral 3,4-dihydroisocoumarins **3** is delivered. The fate of chiral amine byproduct **14** was explored. It was found that some was scavenged by 3,3-dimethyldihydrofuran-2,5-dione to form the amide **15**. And some was decomposed by silver salt possibly via a radical mechanism into benzophenone (**16**) and pyrrolidine that was quickly captured by 3,3-dimethyldihydrofuran-2,5-dione. It should be noted that for the reaction of N-aryl maleimides, dehydrogenation of 3,4-dihydroisocoumarins **3** spontaneously occurs to give isocoumarins **5**. And for the reaction of norbornene, protonation of **III** takes place instead of reductive elimination to deliver hydroarylation products **7**.

Reaction with Aldehydes

Phthalides are of great pharmaceutical[23] and synthetic importance.[24] Many natural products are also phthalides.[23] In the past decades, great efforts have been devoted to their synthesis,[24–25] including their asymmetric synthesis.[24,25b] It is worth noting that transition-metal-catalyzed direct addition of aryl C–H to aldehyde/ketones followed by lactonization emerged as a highly efficient strategy for the construction of various phthalides.[26] Attractively, reactants are simple and readily available. However, the asymmetric variants remain rare. In 2015, Cramer and co-workers[27] reported a chiral rhodium(III)-catalyzed asymmetric intramolecular Grignard-type addition of inert C–H bond to aldehydes followed by annulation to give chiral phthalides (3 examples) in moderate enantioselectivities. In 2019, we[28] reported a highly enantioselective synthesis of chiral phthalides by a chiral transient directing group assisted, rhodium(III) catalyzed Grignard-type addition of arene C–H bond of one aldehyde to the carbonyl group of another followed by annulation. In 2020, we[29] developed an alternative synthesis of chiral 3-aryl substituted phthalides by chiral rhodium(III)-catalyzed Grignard-type addition of inert arene C–H bond of chiral benzamide to aldehyde followed by lactonization. Despite these achievements, two issues remain unresolved. First, previously reported methods turned out to be ineffective to aliphatic aldehydes to give chiral 3-alkyl substituted phthalides, which ubiquitously exist in natural products and bioactive compounds. Second, the previously reported methods are not economical enough for practical uses considering (1) precious rhodium was used as catalyst, (2) the chiral amines to form chiral directing groups as well as the chiral cyclopentadienyl ligands are not facilely available.

Herein, we present a highly diastereoselective ruthenium(II)-catalyzed asymmetric C–H activation of chiral (*R*)-α,α-diphenylprolinol-derived benzamides with aldehydes to afford various chiral phthalides. As shown in Scheme 7, with chiral benzamide as the model substrate, aromatic aldehydes bearing a wealth of functional groups in the *para*-, *ortho*-, or *meta*-position were tested under the optimized reaction conditions, furnishing phthalides in moderate to good yields with high ee values (**19a–h**). Then various benzamides were allowed to react with 4-nitrobenzaldehyde, giving the corresponding phthalides (**19i–o**, **19q–r**) in good yields with high ee values. But the *ortho*-methyl benzamide was unreactive probably due to the steric hindrance. Interestingly, ethyl 3,3,3-trifluoropyruvate could also be successfully converted to the desirable phthalide (**19s**). Notably, aliphatic aldehydes could also be applied,



Scheme 7. Reaction of chiral amides with aldehydes. Reaction conditions: under N_2 atmosphere, amide **1** (0.2 mmol), aldehyde **18** (0.4 mmol), $[RuCl_2(p\text{-cymene})]_2$ (10 mol%), $AgPF_6$ (1 equiv.), $Zn(OAc)_2$ (2 equiv.), THF (0.2 mL), 65 °C, and 48 h. [a] The absolute configuration was assigned to be *S* by comparison of its optical rotation with the reported value (see Supporting Information). [b] amide (2.5 mmol).

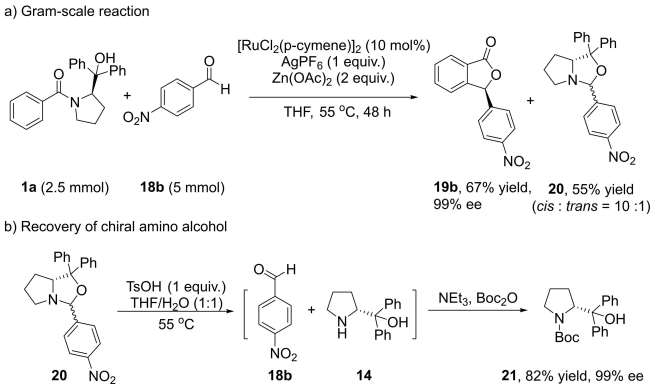
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making it possible to synthesize many valuable chiral 3-alkyl phthalide natural products (Scheme 9).

To check the practicability of this methodology, we attempted to perform the reaction of **1a** and **18b** on a large scale of 2.5 mmol (Scheme 8a). Delightedly, the product **19b** was obtained in 67 % yield with 99 % ee. It is worth noting that the chiral oxazolidine byproduct **20** (*cis*:*trans* = 10:1) was obtained, from which α,α-diphenylprolinol could be facilely recovered through two steps in the form of Boc-protected α,α-diphenylprolinol **21** without loss of any enantiomeric purity (> 99 % ee) (Scheme 8b). This finding also revealed why excessive aldehydes were required for this catalytic reaction.

As shown in Scheme 9, our methodology has been proved to be a highly efficient tool to construct various enantiomerically enriched chiral phthalide natural products and bioactive compounds. First, the Vermistatin[30] analogue **19t** was obtained in 40 % yield with 90 % ee by the reaction of the chiral amide **1k** and the aldehyde **18i** (Scheme 9a). Secondly, with the chiral amide **1u** as the substrate, divergent synthesis of four phthalides **19u–aa** were accomplished in a single step from the corresponding aliphatic aldehydes **18j–m** (Scheme 9b). Full demethylation by BBr_3 gave the natural product Cytosporone E (**19x**),[31] Cytosporone Q (**19y**),[32] **19z** with in vitro growth inhibitory activity,[33] and **19ab** in good yields without loss of enantiomeric excesses. To the best of our knowledge, this is the first asymmetric total synthesis of Cytosporone Q (**19y**). The phthalide **19ab** could be converted into the anti-HIV active compound **19ac**.[34] Similarly, reaction of the chiral amide **1q** with the aldehyde **18n** led to a straightforward synthesis of the anti-inflammatory and neuroprotective molecule **19ad** (Scheme 9c).[35] Reaction of the chiral amide **1v** with the aldehyde **18n** gave a mixture of regioisomers **19ae** and **19ae′** (5:1, 38 % overall yield). Although it is difficult to separate them by column chromatography, pure (R)-Chuangxinol (**19ae**)[36] could be obtained with > 99 % ee by a single recrystallization (75 % yield, Scheme 9d). Additionally, (*S*)-Concentricolide (**19af**) was successfully synthesized from the chiral amide **1w** and propionaldehyde (**18o**) in 45 % yield with 92 % ee (Scheme 9e), which is much shorter than the reported synthetic routes.[37] Moreover, with the chiral amide **1x** prepared from 3-methoxybenzaldehyde via 4 steps as the substrate, Eleutherol (**19ag**)[38] could be facilely synthesized in 35 % yield with 90 % ee. Notably, although racemic Eleutherol has been previously synthesized via 5—10 steps,[39] this work represents the first asymmetric synthesis of Eleutherol. It should be mentioned that Eleutherol is a key intermediate to synthesize Eleuthoside A (**19ah**, Scheme 9f).[39a]



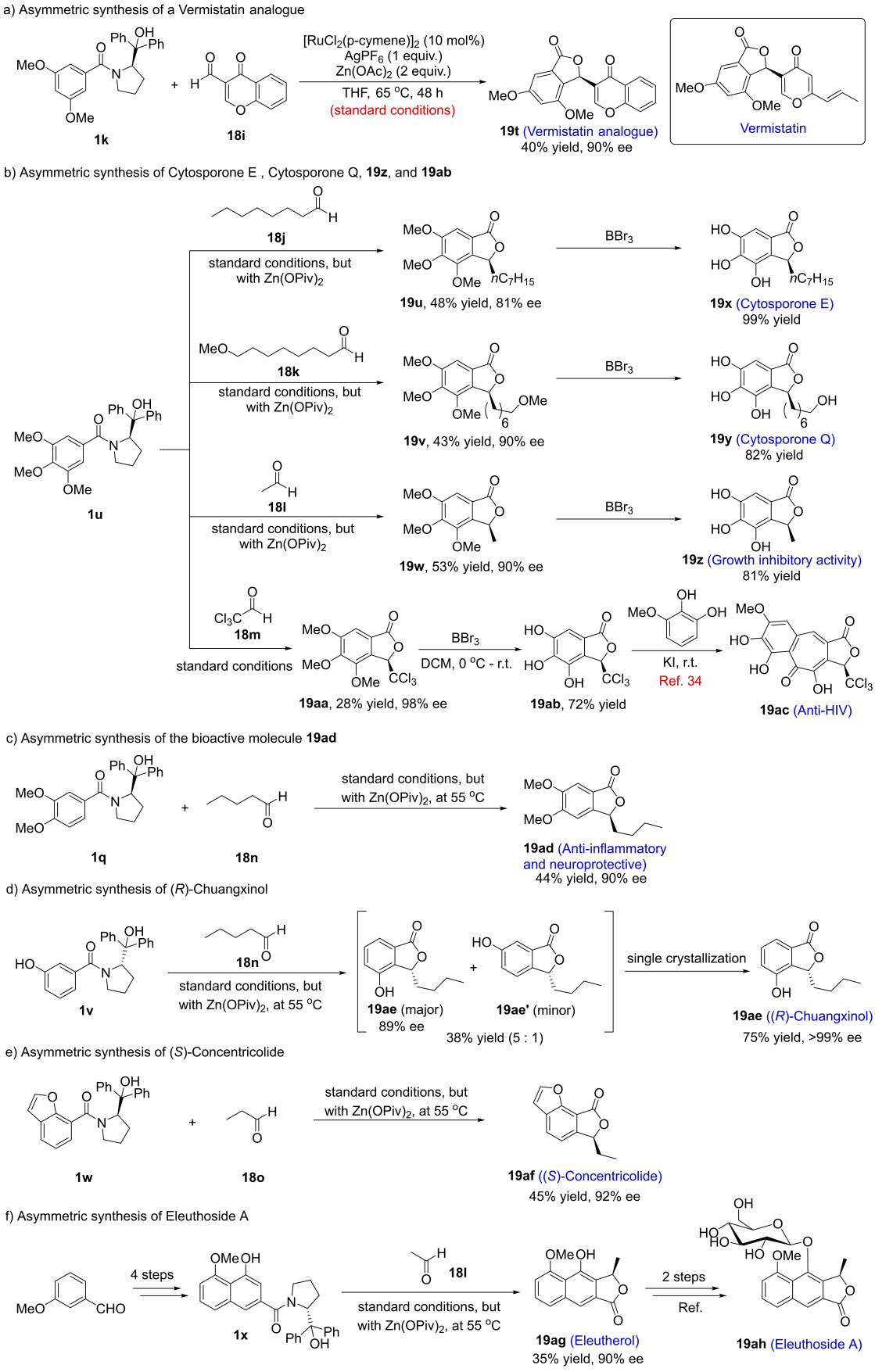
Scheme 8. Gram-scale reaction and recovery of chiral amino alcohol.

To shed some light on the reaction mechanism, some control experiments were conducted. First, it was found chiral bisarene ruthenium complexes **Ru-A** and **Ru-B** were stereoselectively formed in high yields when the chiral amide **1a** and **1d** were respectively treated with $[Ru(p\text{-cymene})Cl_2]_2$ and $AgPF_6$ (Scheme 10a). Under standard catalytic reaction conditions, p-cymeneRu^{II} also reacted with the chiral amide **1a** to form **Ru-A** quantitatively as determined by ^1H NMR (Scheme 10b). If the aldehyde **18b** was introduced and the reaction was allowed to continue to run for another 36 h, the product **19b** was obtained in 55 % yield with 99 % ee (Scheme 10c). In addition, when $[Ru(p\text{-cymene})Cl_2]_2$ used in standard reaction conditions was replaced with **Ru-A**, the reaction of **1a** with **18b** could still proceed smoothly to give the product **19b** in 51 % yield with 99 % ee (Scheme 10c). Therefore, **Ru-A** should be the precatalyst. Next, to further identify the real catalytically active species, cross-scrambling experiments were conducted (Scheme 10d). It was observed that the combined use of the chiral amide **1d** with excessive **Ru-A** (50 mol %) only led to the product **19l** (67 % yield, 92 % ee), while the combined use of the chiral amide **1a** and excessive **Ru-B** (50 mol %) only provided the product **19b** (51 % yield, 99 % ee). It implied that the catalytically active ruthenium species was generated from **Ru-A** or **Ru-B** by the dissociation of p-cymene rather than the chiral amide. Finally, chiral-matching studies were carried out, considering both the amide substrate and the ruthenium catalyst were chiral in this reaction (Scheme 10e). By taking the reaction of the amide **1a** and the aldehyde **18b** as the benchmark reaction, in the presence of the precatalyst **Ru-B**, it was found that the (*R*)-amide substrate led to the product with 99 % ee, whereas the (*S*)-amide substrate led to the product with −99 % ee (with the opposite absolute configuration). The two reactions are in similar yields (52 % and 61 %). When the racemic amide was used, the racemic product was obtained in 55 % yield. Accordingly, the stereoselectivity of this reaction should be exclusively controlled by the chiral amide.

Based on the above results and previous studies,[27–29] a plausible reaction mechanism is proposed (Scheme 11). First, in the presence of $AgPF_6$, the chiral amide **1a** reacts with $[Ru(p\text{-cymene})Cl_2]_2$ to form the bisarene ruthenium complex **Ru-A**. Upon the dissociation of *p*-cymene, the catalytically active species **Ru-A′** was generated. By the assistance of acetate, the chiral amide substrate **1a** reacts with **Ru-A′** to form the intermediate **I** via the CMD mechanism. Addition of C–Ru bond to the aldehyde **18** affords the intermediate **II**, which further cyclizes upon the amide to give the ruthenium alkoxide intermediate **III**. Transmetallation with zinc cation regenerates the catalyst and forms the Zn alkoxide **IV**, which collapses upon

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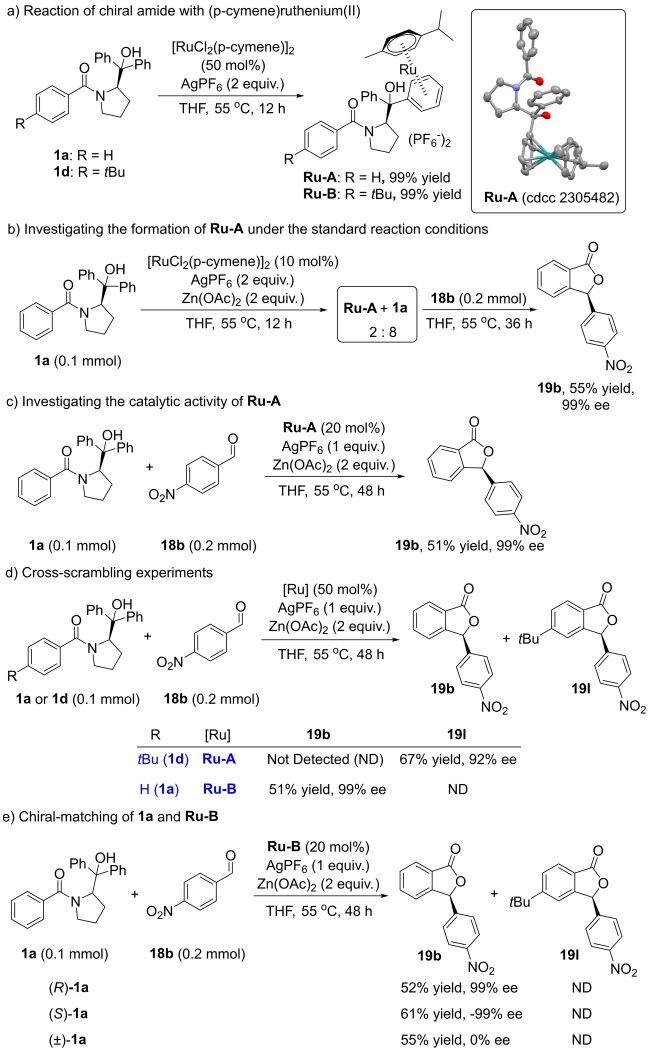
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Scheme 9. Synthesis of chiral phthalide natural products and bioactive compounds.

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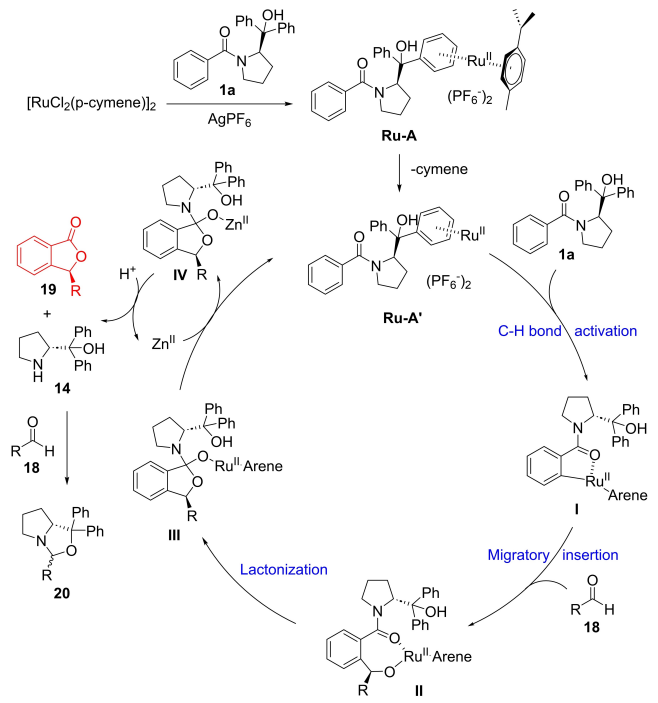


Scheme 10. Control experiments for Ru-catalyzed asymmetric acylarylation of aldehydes.

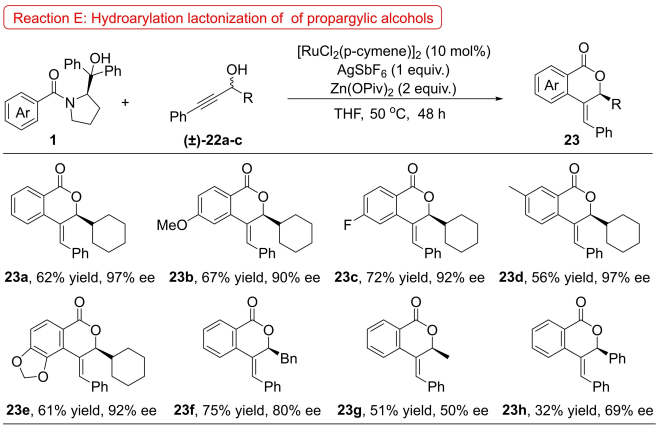
protonation to release the product **19** and the chiral amine **14**. Reaction of the chiral amine **14** with aldehyde generates the chiral oxazolidine byproduct **20**.

Reaction with Propargylic Alcohols

Inspired by the work by Li[40] and Gooßen[41] on the synthesis of 4-benzylidene-3,4-dihydroisocoumarin from amide or benzoic acid with propargylic alcohols by C–H activation, we also attempted the reaction of the chiral amide **1a** with 2.0 equivalent of the racemic secondary propargylic alcohol **22a** (Scheme 12). It was found that the chiral amides preferably reacted with one enantiomer of the propargylic alcohol **22a** via a C–H hydroarylation/lactonization sequence, giving the 4-benzylidene-3,4-dihydroisocoumarins **23a** in 38 % yields with 97 % ee. Surprisingly, the alcohol **22a** was recovered in 33 % yield with 0 % ee, suggesting its racemization occurred quickly under reaction conditions. When 3.0 equivalent of the racemic secondary propargylic



Scheme 11. Proposed reaction mechanisms for Ru-catalyzed asymmetric addition of inert arene C–H bond to aldehydes.



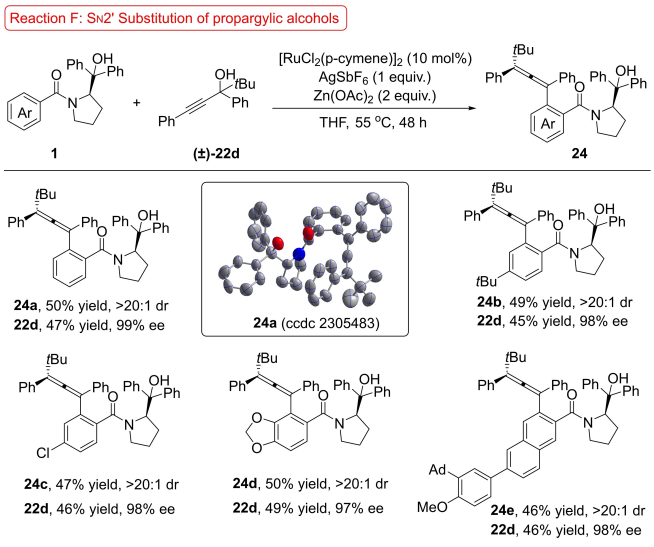
Scheme 12. Reaction of chiral amides with secondary propargylic alcohols. Reaction conditions: under N_2 atmosphere, amide **1** (0.1 mmol), propargylic alcohol **22** (0.3 mmol), $[RuCl_2(p\text{-cymene})]_2$ (10 mol %), $AgSbF_6$ (1 equiv.), $Zn(OPiv)_2$ (2 equiv.), THF (0.1 mL), 50 °C, and 48 h.

alcohol **22a** was used, the product **23a** was obtained in an improved yield of 62 % with 97 % ee. Chiral benzamides with electron-donating or electron-withdrawing groups on the *para*- and *ortho*-positions were tolerable (**23b–e**). The enantioselectivity of the reaction decreased along the decrease of the steric bulk of the R substituent of propargylic alcohols (**23f–g**). Both the yield and ee value decreased if R was phenyl (**23h**).

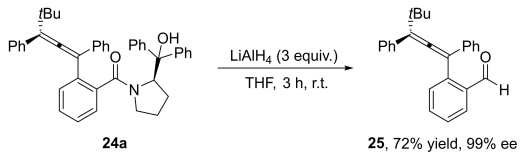
Interestingly, in accord with the previous studies,[42] when the racemic tertiary propargylic alcohol **22d** was employed, the axially chiral allene product **24a** was obtained in 50 % yield with

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Scheme 13. Reaction of chiral amide **1** with tertiary propargylic alcohols **22d**. Reaction conditions: under N_2 atmosphere, amide (0.15 mmol), propargylic alcohol (0.2 mmol), $[RuCl_2(p\text{-cymene})]_2$ (10 mol%), $AgSbF_6$ (1 equiv.), $Zn(OAc)_2$ (2 equiv.), THF (0.1 mL), 55 °C, 48 h.



Scheme 14. Removal of chiral DG from the allene product **24a**.

>20:1 dr (Scheme 13). Besides, the propargylic alcohol **22d** was recovered in 47% yield with 99% ee. Accordingly, the chiral amides only reacted with one enantiomer of the propargylic alcohol, leaving the other intact. Similarly, several substituted benzamides were also converted into allenylated products **24b–d** in good yields with excellent diastereoselectivity (>20:1 dr). In addition, the Adapalene derived allene **24e** could be prepared as well. Notably, in all the cases the propargylic alcohol **22d** was recovered in 45–46% yield with 97–98% ee. The chiral allenylated benzaimde **24a** could be converted into the structrually interesting chiral allenylated benzaldehyde **25** without loss of enantiomeric purity by $LiAlH_4$ reduction of the chiral amido group at room temperature, which proceeded analogueously to the reduction of Weinreb amides (Scheme 14).

Conclusion

In summary, we describe a versatile and readily available chiral amide directing group for ruthenium-catalyzed asymmetric C–H bond activation of amides with olefins, aldehydes, and propargylic alcohols. A series of chiral products including 3,4-dihydroisocoumarins, isocoumarins, phthalides, chiral bicyclo[2.2.1]heptanes, 4-alkylidene-3,4-dihydroisocoumarins, and allenes were prepared in good yields with high stereoselectivities. Moreover, asymmetric syntheses of many natural products and bioactive compounds, such as Montroumarin, Cyclosporone E, Cyclosporone Q, Concentricolide, Chuangxinol, and Eleutherol, have been achieved very concisely by using this methodology as the key step.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Asymmetric C–H Activation · Chiral Directing Group · Dihydroisocoumarin · Phthalide · Ruthenium(II)

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[43] Deposition numbers 2313647 (for **5c**), 2305481 (for **7e**), 2305482 (for **Ru-A**), 2305483 (for **24a**) contain the supplementary crystallographic data for this paper. These data are provided free of charge by the joint Cambridge Crystallographic Data Centre and Fachinformationszentrum Karlsruhe Access Structures service.

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